A Machine Learning-Driven Stray Animal Welfare Intelligence Framework for Bangladesh

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***Abstract*—Stray animal welfare in Bangladesh is closely connected with public health, human–animal conflict, disease prevention, and humane urban animal management. However, relevant information is fragmented across online news reports and localized field observations, limiting the ability to obtain structured and actionable welfare intelligence. This study presents a machine learning-driven, dual-source stray animal welfare intelligence framework for Bangladesh that integrates Bangladesh-focused online news analysis with a pilot field survey conducted in Badda, Dhaka.**

The news intelligence pipeline initially collected 1,094 records and applied relevance filtering, geographic scope refinement, weak labeling, and assisted review. This process produced a final machine learning corpus of 226 Bangladesh-focused records. TF–IDF text representations were evaluated using Linear Support Vector Machine (SVM), Logistic Regression, and Complement Naive Bayes classifiers. Linear SVM achieved the best overall performance, with a macro F1-score of 0.6068 and a weighted F1-score of 0.7517. An interpretable Stray Abuse Risk Index (SARI V2) was additionally used as a separate heuristic assessment mechanism, producing a mean score of 1.550 and a maximum score of 5.500.

To complement media-derived intelligence with ground-level evidence, a structured pilot survey of 325 stray animals was conducted across selected areas of Badda, Dhaka. The survey recorded animal type, age, observed health condition, vaccination status, sterilization status, behavior, and location. Within the surveyed sample, 9.85% of animals were reported as vaccinated, 10.46% as sterilized, and 28.62% showed observed nutritional health concerns. The field survey is treated as a localized case study rather than a nationally representative prevalence estimate.

By combining broader Bangladesh-focused incident intelligence with localized field observations, the proposed framework demonstrates a scalable approach for transforming heterogeneous animal-welfare information into structured evidence. The framework may support future data-driven vaccination, sterilization, welfare monitoring, and intervention planning while providing a foundation for expansion to additional areas of Bangladesh.

***Index Terms*—stray animal welfare, machine learning, natural language processing, human–animal conflict, rabies, SARI, field survey, Bangladesh, One Health, decision support**

I. Introduction

Free-roaming dogs and cats are a visible component of urban environments in Bangladesh, where animal welfare, public health, community safety, and human behavior are closely interconnected. Stray animals may experience food scarcity, malnutrition, disease, injury, abuse, poisoning, and inadequate access to veterinary care. At the same time, concerns regarding dog bites, rabies transmission, aggressive behavior, and unmanaged animal populations can contribute to negative human–animal interactions. Consequently, the management of free-roaming animals is not solely an animal-welfare issue; it also represents a public-health and community-management challenge.

Rabies provides a particularly important example of this relationship. Bangladesh has implemented large-scale dog vaccination and rabies-control programs, demonstrating the importance of coordinated interventions involving human health, animal health, and local communities. Such challenges are consistent with the One Health perspective, in which human, animal, and environmental health are considered interconnected. Nevertheless, vaccination and disease control represent only part of the broader stray-animal welfare problem. Abuse, killing, poisoning, rescue events, nutritional condition, sterilization coverage, and human–animal conflict also require systematic monitoring.

A major challenge is the fragmentation of relevant information. Reports of animal abuse, rabies concerns, dog bites, rescue operations, poisoning, and other welfare-related events are distributed across numerous online news sources. These reports are primarily unstructured text and are therefore difficult to analyze systematically at scale. Field-level information creates a second challenge. Variables such as visible health condition, vaccination, sterilization, age, and animal behavior are generally unavailable from news reports alone. Consequently, relying on either media reports or localized field observations in isolation provides only a partial view of the problem.

An earlier version of this research established a news-based analytical framework for Bangladesh in which online reports were collected, processed, categorized using rule-based techniques, and evaluated using a custom Stray Abuse Risk Index (SARI). Although the framework demonstrated the feasibility of transforming dispersed news reports into structured incident information, several limitations remained. Classification depended primarily on keyword-based rules, many headlines contained insufficient contextual information, geographic information was inconsistent, and the system lacked direct field-level observations of stray-animal welfare conditions.

The present study extends that foundation by introducing a machine learning-driven dual-source welfare intelligence

framework. The first component, termed *News Intelligence*, collects and refines Bangladesh-focused news records and applies natural language processing and supervised text classification to identify incident categories. The second component, termed *Field Intelligence*, incorporates structured ground-level observations collected through a pilot survey in Badda, Dhaka. Rather than merging these heterogeneous datasets at the individual-record level, the framework treats them as complementary evidence streams.

The news-processing pipeline was designed with multiple refinement stages. A total of 1,094 news records were initially assembled. Relevance filtering retained 690 records, after which geographic scope filtering identified 250 Bangladesh-primary records. Assisted review and corpus preparation produced a final set of 226 records for machine-learning experiments. Three traditional text classifiers—Linear Support Vector Machine (SVM), Logistic Regression, and Complement Naive Bayes—were evaluated using TF–IDF representations. Linear SVM achieved the highest macro F1-score among the evaluated models.

A revised SARI mechanism, referred to as SARI V2, was maintained separately from the machine-learning classification target. Instead of training a model to reproduce a risk label generated from deterministic rules, SARI V2 serves as an interpretable heuristic index based on textual indicators associated with abuse, human conflict, sickness or rabies, poisoning, and contextual incident severity. This separation reduces circularity between incident classification and risk assessment while preserving interpretability.

To introduce direct field evidence, 325 stray animals were surveyed in four localities within Badda, Dhaka: Uttar Badda, Dokkhin Badda, Moddho Badda, and Jheelpar. The survey captured information on animal type, age, approximate weight, observed health condition, visible health issues, vaccination, sterilization, behavior, and location. Badda is used as a pilot case-study area for demonstrating the field-intelligence component of the proposed framework. The observations are therefore not interpreted as nationally representative estimates of stray-animal conditions in Bangladesh.

The main contributions of this study are summarized as follows:

- A Bangladesh-focused, dual-source stray-animal welfare intelligence framework integrating online news analysis with localized field evidence.
- A multi-stage data-refinement pipeline that reduces a heterogeneous collection of 1,094 news records to a more focused 226-record corpus for incident-classification experiments.
- A comparative evaluation of Linear SVM, Logistic Regression, and Complement Naive Bayes using TF–IDF text features and class-sensitive evaluation metrics.
- An interpretable SARI V2 mechanism maintained separately from the supervised incident-classification target.
- A structured pilot field survey of 325 stray animals in Badda, Dhaka, providing localized evidence regarding observed health, vaccination, sterilization, behavior, and spatial variation.
- An evidence-level integration strategy that combines media-derived incident intelligence and field observations without incorrectly treating the two heterogeneous datasets as directly interchangeable samples.

The resulting framework is intended as a scalable analytical foundation rather than a complete national surveillance system. By extending field data collection to additional areas and improving the quality and coverage of textual incident data, the proposed approach can support broader animal-welfare monitoring and intervention planning in Bangladesh.

II. Literature Review and State of the Art

A. Stray Animal Management and Rabies in Bangladesh

Free-roaming animal management in Bangladesh has historically been closely associated with rabies prevention and public-health interventions. Rabies is a vaccine-preventable zoonotic disease, and dogs are the principal source of human rabies transmission in endemic settings. Consequently, dog vaccination, population management, bite prevention, and access to post-exposure prophylaxis are central components of rabies-control strategies [1].

Bangladesh has progressively adopted a One Health-oriented approach to rabies control. The national strategy has included mass dog vaccination (MDV), post-exposure prophylaxis for bite victims, public awareness, and multisectoral coordination. According to the World Health Organization, approximately 1.7 million dogs live in Bangladesh, with a large proportion free-roaming. Bangladesh expanded its mass dog vaccination activities from an initial pilot intervention into a broader national programme [2].

Tenzin et al. conducted an important field-based investigation of free-roaming dogs in Dhaka City [3]. Using surveys across sampled city wards, the study estimated the free-roaming dog population and evaluated the progress of a Catch-Neuter-Vaccinate-Release (CNVR) programme. The work demonstrated the importance of localized field observation for understanding animal populations and evaluating vaccination and sterilization interventions. However, its primary objective was population estimation and rabies-control programme assessment rather than automated analysis of animal-welfare incidents from heterogeneous information sources.

More recently, Ghosh et al. analyzed Bangladesh's rabies-control programme using data spanning 2011–2023 [4]. Their One Health study integrated multiple epidemiological and vaccination-related data sources and reported an estimated national dog population of approximately 1.67 million. The analysis indicated that expansion of mass dog vaccination and human anti-rabies vaccination was associated with substantial progress in rabies control. Such findings demonstrate the value of combining multiple data sources for animal and public-health decision making.

Recent field evidence further demonstrates the importance of localized vaccination monitoring in Bangladesh. Swedberg

et al. evaluated mass dog vaccination strategies across four urban and peri-urban study sites and reported that 9,195 dogs were vaccinated during the evaluated campaigns [5]. Post-vaccination surveys indicated that more than 93% of observed dogs were free-roaming, while estimated vaccination coverage varied substantially depending on both the vaccination strategy and the enumeration method. The study highlights the practical importance of location-specific monitoring and reliable field-survey methods rather than relying on a single national population assumption.

These studies establish a strong foundation for rabies surveillance and dog population management in Bangladesh. Nevertheless, rabies represents only one dimension of stray-animal welfare. Abuse, killing, poisoning, rescue events, nutritional condition, visible health problems, sterilization status, and human–animal conflict require broader analytical approaches.

B. News-Based Event Intelligence and Text Classification

Online news platforms provide a continuously growing source of unstructured information about social and public-health events. News reports can contain information regarding animal abuse, poisoning, killing, rescue operations, rabies concerns, bite incidents, community responses, and animal-management activities. However, manually reviewing large numbers of reports is time-consuming and makes systematic monitoring difficult.

Natural language processing (NLP) and machine-learning techniques provide a mechanism for transforming such unstructured text into structured event categories. In a conventional text-classification pipeline, textual documents are first converted into numerical representations before being supplied to a classifier. Term Frequency–Inverse Document Frequency (TF–IDF) remains a widely used representation for text retrieval and classification because it emphasizes terms that are informative within a document while reducing the influence of terms that occur frequently across the corpus [6].

Linear classifiers are particularly suitable for sparse, high-dimensional text representations. Support Vector Machines have been widely studied for such classification problems [7], and Joachims demonstrated their effectiveness specifically for text categorization with many relevant features [8]. Complement Naive Bayes provides another useful approach for addressing limitations of conventional Naive Bayes assumptions in text classification [9]. Accordingly, the present study evaluates Linear SVM, Logistic Regression, and Complement Naive Bayes using the same refined news corpus rather than assuming that a more computationally complex model will necessarily perform better.

Model evaluation also requires more than overall accuracy when the class distribution is imbalanced. Macro-averaged precision, recall, and F1 score provide a useful view of performance because each class contributes equally to the macro average, including classes with relatively few observations [10]. The machine-learning pipeline was implemented using Scikit-learn [11]. In addition to predictive performance, feature-level inspection was used to examine which textual terms contributed most strongly to individual class decisions, following the broader importance of feature analysis in text-classification systems [12].

C. Weak Supervision and Assisted Label Construction

A major limitation in domain-specific text classification is the cost of constructing a sufficiently large manually annotated training dataset. Weak-supervision approaches address this problem by using heuristics, rules, existing knowledge, or other imperfect sources to generate candidate labels.

Ratner et al. introduced Snorkel as a framework for creating training data from user-defined labeling functions instead of relying exclusively on manually labeled examples [13]. Their work demonstrates the broader principle that imperfect programmatic supervision can be useful when large manually labeled datasets are unavailable.

The labeling procedure in the present study is inspired by this general weak-supervision principle but does not implement the Snorkel system. Candidate incident labels are first generated through domain-specific textual rules. A stratified validation sample is subsequently inspected through an assisted review process to identify obvious out-of-scope records, ambiguous cases, and incorrect suggested labels. The resulting labels should therefore be interpreted as weakly supervised and partially reviewed labels rather than as a fully independent expert-annotated gold-standard corpus.

This distinction is particularly important when interpreting classifier performance. Cross-validation scores in this study measure predictive agreement with the constructed labels and should not be interpreted as an independent estimate of expert-level classification accuracy.

D. Field-Based Welfare Observation

Reliable management of free-roaming animal populations also depends on appropriate field-observation and population-estimation methods. Belo et al. systematically reviewed approaches used to estimate free-ranging dog populations and emphasized the importance of understanding population structure and abundance for the planning and monitoring of population-control and animal-welfare interventions [14]. This broader methodological perspective supports the use of localized field evidence alongside event-oriented information sources.

Evidence from dog population management research also shows that intervention effectiveness cannot be assessed using population size alone. Smith et al. systematically reviewed dog population management studies across multiple countries and found that programmes have been evaluated using outcomes related to population size, dog health and welfare, public-health risk, and community attitudes [15]. Their review also identified substantial variation in study design and reporting quality, reinforcing the need for clearly defined indicators when evaluating vaccination, sterilization, and other management interventions.

More recent experimental evidence also shows why intervention outcomes should be evaluated rather than assumed. Fielding et al. conducted a multi-site randomised controlled trial of intensive surgical sterilisation in free-roaming dog populations [16]. Female sterilisation coverage reached approximately 58–66% in intervention areas. Although reductions were observed in puppies, lactating females, and some reported nuisance behaviours, the study did not find a significant reduction in adult free-roaming dog counts during the two-year follow-up. These findings illustrate that sterilisation outcomes can depend on population movement, coverage, and local conditions, reinforcing the need for location-specific monitoring alongside intervention delivery.

News-derived information provides evidence regarding events that receive media coverage, but it cannot reliably describe the everyday welfare condition of individual free-roaming animals. Variables such as observed nutritional condition, approximate age, vaccination status, sterilization status, visible health problems, and behavior generally require direct field-level observation.

Localized animal surveys therefore provide a complementary evidence source. In Dhaka, Tenzin et al. used direct field observation not only to estimate free-roaming dog populations and CNVR coverage but also to record observable characteristics including approximate age, body condition, skin condition, and sterilization and vaccination status [3]. Their findings illustrate how structured field observations can provide information that is not normally available from event-oriented news reports.

Building on this field-based perspective, the present study incorporates a structured survey of 325 observed animals from four localities in Badda, Dhaka. The survey records variables related to animal type, approximate age, observed nutritional condition, vaccination status, sterilization status, behavior, visible health concerns, and location. Unlike a national prevalence survey, this component is treated as a localized pilot case study. Its purpose is to demonstrate how field-level welfare observations can complement the Bangladesh-focused news intelligence pipeline without implying that the Badda sample represents the national stray-animal population.

E. Need for Multi-Source Welfare Intelligence

International guidance increasingly treats free-roaming dog management as a multisectoral problem involving animal welfare, public health, population management, community behavior, and disease prevention. The World Organisation for Animal Health (WOAH) recommends a holistic dog population management approach that improves dog health and welfare while addressing public-health and safety concerns. Importantly, such programmes should consider local dog population dynamics and community conditions rather than applying identical interventions across different settings [17].

This perspective is consistent with the broader One Health framework. The global "Zero by 30" strategy emphasizes coordinated surveillance, mass dog vaccination, community engagement, and access to human post-exposure prophylaxis for eliminating dog-mediated human rabies deaths [18]. More broadly, the Quadripartite One Health Joint Plan of Action promotes collaboration across human, animal, and environmental health systems for monitoring and responding to interconnected health risks [19]. These principles support the development of analytical systems that combine multiple forms of evidence rather than treating animal welfare and public-health observations as isolated problems.

Existing Bangladesh-focused research provides valuable evidence regarding rabies epidemiology, vaccination, and free-roaming dog population management. However, these objectives differ from the problem of building an integrated animal-welfare intelligence system.

A broader system should be capable of examining both reported incidents and ground-level welfare indicators. News reports provide one perspective by capturing socially visible incidents, whereas field surveys provide another by documenting observed animal conditions within a defined locality. These sources have different units of observation and sampling mechanisms and therefore should not be directly merged as if they represented a single homogeneous dataset.

The present study addresses this issue through a dual-source architecture. Bangladesh-focused news records are processed through an NLP and machine-learning pipeline to create incident intelligence, while the Badda pilot survey provides localized field intelligence. The two components are integrated at the evidence and decision-support level rather than through row-level data fusion.

Recent evidence also highlights continuing methodological gaps in the monitoring of free-roaming dog populations. A scoping review by Cunha Silva et al. found substantial variation in dog-enumeration and movement-study methods across rabies-endemic settings and identified important gaps in the evidence needed for effective intervention planning [20]. These limitations further support the use of clearly defined, location-aware indicators when developing welfare and public-health decision-support systems.

III. Research Gap and Contributions

Despite substantial progress in rabies control and free-roaming dog management, several gaps remain in the development of data-driven stray-animal welfare intelligence for Bangladesh.

First, much of the existing Bangladesh-focused literature emphasizes rabies, dog population estimation, vaccination, and disease prevention. These are essential public-health objectives, but they do not provide a unified mechanism for monitoring heterogeneous welfare-related incidents such as abuse, killing, poisoning, rescue, disease-related events, and human–animal conflict.

Second, animal-welfare information reported by online media remains unstructured and distributed across multiple sources. A simple keyword-based system can provide an initial analytical solution but has limited ability to generalize across linguistic variations and contextual expressions. This motivates

the transition from purely rule-based incident assignment toward supervised text classification.

Third, the absence of a large expert-annotated Bangladesh-specific stray-animal incident corpus creates a training-data challenge. The present study therefore investigates a weak-supervision-inspired labeling strategy combined with assisted review while explicitly acknowledging that the resulting corpus is not equivalent to independently annotated ground truth.

Fourth, news reports alone cannot provide sufficient information regarding the routine condition of free-roaming animals. Media coverage is event-driven and may preferentially represent unusual, controversial, or severe incidents. Direct field observations are therefore necessary to provide complementary information regarding health condition, vaccination, sterilization, age, and behavior.

Finally, localized field observations cannot automatically be generalized to the entire country. A framework is needed that can preserve the distinction between broader Bangladesh-focused incident intelligence and localized ground-level evidence while still allowing both to contribute to decision-support.

To address these gaps, this study makes four principal methodological advances over the earlier framework:

1) It replaces exclusive dependence on rule-based incident classification with a comparative machine-learning experiment using TF–IDF features and three supervised text classifiers.
2) It introduces a multi-stage relevance and geographic-scope refinement process to construct a more focused Bangladesh-primary news corpus.
3) It separates incident prediction from the SARI V2 heuristic risk assessment, avoiding the use of deterministic risk labels as a supervised machine-learning target.
4) It extends the framework beyond media intelligence by incorporating a 325-animal pilot field survey in Badda, Dhaka, while preserving the localized nature of the survey findings.

Together, these contributions transform the earlier news-based analytical pipeline into a dual-source welfare intelligence framework that combines machine-learning-based incident analysis with structured field evidence.

IV. Dataset Construction and Data Refinement

The proposed framework relies on two fundamentally different sources of evidence. The first is a collection of online news records used to study reported stray-animal incidents in the Bangladesh context. The second is a structured field survey designed to observe the condition of free-roaming animals within a selected pilot area in Badda, Dhaka. These datasets were kept separate throughout preprocessing and analysis because they represent different units of observation: a news record describes a reported event, whereas a survey record represents an observed animal.

A. News Data Collection

The news component was developed by expanding the data-collection process used in the earlier version of the project. Google News RSS was used as the primary discovery mechanism, with search queries covering topics such as stray dogs and cats, animal abuse, killing, poisoning, rescue, dog bites, rabies, vaccination, disease, and human–animal conflict. Bangladesh-related terms and selected location-oriented queries were also used to improve the coverage of locally relevant reports.

For each retrieved item, the pipeline retained the available headline, source, publication date, link, search context, and RSS description when present. RSS descriptions were useful because headlines alone frequently contained insufficient context for distinguishing between superficially similar incidents.

After combining newly retrieved records with the earlier collection and removing duplicate entries, the expanded dataset contained 1,094 unique news records. Publication dates were successfully parsed for all 1,094 records, and the collection represented 260 unique news sources. Usable RSS descriptions were available for 387 records. The remaining records therefore depended more heavily on information available in their headlines.

This expanded collection was intentionally treated as a broad candidate pool rather than as the final Bangladesh incident dataset. Search-engine retrieval can return records that contain relevant keywords while describing unrelated, foreign, wildlife-related, or otherwise out-of-scope events. Consequently, several refinement stages were applied before the data were used for machine-learning experiments.

B. Content-Based Relevance Filtering

The first refinement stage evaluated whether a retrieved record actually contained evidence of a stray-animal or closely related welfare event. Importantly, relevance was determined from the available title and RSS description rather than from the search query that retrieved the article. This distinction reduced the possibility of retaining a record simply because a relevant phrase had been used during retrieval.

The relevance rules considered textual evidence associated with dogs, cats, stray or free-roaming animals, rabies, animal attacks, abuse, poisoning, killing, rescue, vaccination, disease, and related welfare or conflict concepts. Records without sufficient evidence were moved to a separate rejected collection rather than silently discarded.

From the initial 1,094 records, 690 were retained after content-based relevance filtering, corresponding to a retention rate of approximately 63.1%. A total of 404 records were rejected at this stage. Among the 690 retained records, 256 contained usable RSS descriptions. Dog-related records formed the largest animal category, followed by records in which the animal could not be confidently identified from the available text.

TABLE I
NEWS DATASET REFINEMENT STAGES

Stage	Records	Retained
Expanded candidate collection	1,094	100.0%
Content-relevant records	690	63.1%
Bangladesh-primary records	250	22.9%
Final ML corpus	226	20.7%

C. Bangladesh Scope Refinement

Relevance to stray animals does not necessarily imply relevance to Bangladesh. The initial relevance-filtered corpus still contained reports from other countries as well as records whose geographic relationship with Bangladesh was uncertain. A second filtering stage was therefore introduced to separate the data according to geographic and topical scope.

Geographic evidence was inferred from explicit references to Bangladesh, recognized Bangladeshi locations, and a curated set of Bangladesh-based news sources. The search query itself was deliberately excluded as evidence of geographic origin because a Bangladesh-oriented query can still retrieve a foreign event.

The scope-refinement procedure separated the 690 relevant records into three groups:

- **Bangladesh-primary**: records with sufficient evidence that the event or reporting context was relevant to Bangladesh;
- **Foreign/background**: relevant animal-welfare records that could provide contextual information but did not describe a Bangladesh-primary event; and
- **Out-of-scope**: records dominated by wildlife, unrelated meanings, or other content unsuitable for the target incident corpus.

This procedure produced 250 Bangladesh-primary records, 331 foreign/background records, and 109 out-of-scope records. Thus, only 36.2% of the 690 relevance-filtered records were retained as Bangladesh-primary evidence. This reduction illustrates why keyword retrieval alone was insufficient for constructing the final analytical corpus.

Among the 250 Bangladesh-primary records, 175 contained explicit Bangladesh or local-place evidence, while another 75 were retained through evidence associated with Bangladesh-based sources. The records covered the period from 2011 to 2026 and represented 48 unique primary news sources.

Figure 1 summarizes the complete refinement process.

D. Initial Incident Label Construction

The Bangladesh-primary corpus did not contain manually supplied incident labels. Initial labels were therefore generated using a weak-supervision-inspired rule system based on textual indicators in the available headline and description.

Eight incident categories were considered: *Abuse*, *Conflict*, *Disease*, *Human-Animal Conflict*, *Killing*, *Other*, *Poisoning*, and *Rescue*. These categories were selected to represent both direct animal-welfare incidents and broader events relevant to interactions between free-roaming animals and human communities.

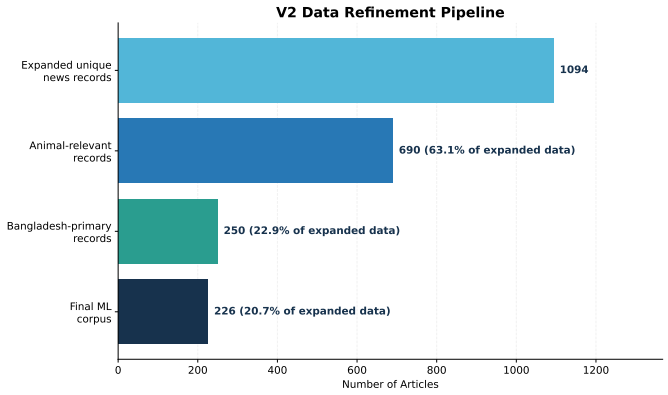


Fig. 1. Refinement of the news corpus from the expanded candidate collection to the final machine-learning dataset.

The initial labeling process assigned the 250 Bangladesh-primary records as follows: 99 Other, 39 Disease, 35 Killing, 28 Conflict, 26 Abuse, 13 Human-Animal Conflict, 7 Rescue, and 3 Poisoning. The strong imbalance between categories was retained rather than artificially balancing the dataset through synthetic observations, because the objective was to preserve the characteristics of the collected corpus.

E. Assisted Review and Corpus Finalization

Programmatically generated labels are imperfect. A stratified validation sample of 64 records was therefore constructed to inspect the behavior of the labeling procedure across incident categories. Each sampled record was examined using its available title and description and assigned one of three review decisions: *Keep*, *Exclude*, or *Ambiguous*.

The review retained 40 records, excluded 22, and marked 2 as ambiguous. Examples of excluded content included wildlife-related events, foreign incidents, unrelated uses of animal terminology, political expressions such as “dog whistle,” and records that did not describe a relevant incident. Among the 40 retained records, the initial suggested label agreed with the assisted-review label in 90.0% of cases.

This 90.0% agreement should not be interpreted as classifier accuracy. It measures agreement between the programmatically suggested labels and the assisted review for the subset of records that were retained. Furthermore, the complete corpus was not independently annotated by multiple domain experts. The labels used for machine learning are therefore described throughout this paper as weakly supervised and partially reviewed rather than as a gold-standard annotation set.

After removing reviewed exclusions and ambiguous cases from the Bangladesh-primary collection, 226 records remained in the final machine-learning corpus.

The final distribution is highly imbalanced. In particular, Poisoning and Rescue contain only three and two records, respectively. This characteristic has direct implications for model evaluation: performance estimates for the rarest classes

TABLE II
FINAL MACHINE-LEARNING CORPUS BY INCIDENT CATEGORY

Incident Category	Records
Other	90
Disease	39
Killing	31
Conflict	29
Abuse	21
Human-Animal Conflict	11
Poisoning	3
Rescue	2
Total	**226**

are inherently unstable, and overall accuracy alone would provide an incomplete description of classifier behavior.

F. Pilot Field Survey Dataset

The second evidence stream was constructed independently from the news corpus. A structured field survey recorded 325 free-roaming animals across four localities in Badda, Dhaka: Uttar Badda, Dokkhin Badda, Moddho Badda, and Jheelpar.

The survey was designed to capture variables that cannot normally be inferred reliably from news reports. These included animal type, estimated age group, approximate weight, observed nutritional or general health condition, visible health issues, vaccination status, sterilization status, behavior toward humans, and the location in which the animal was observed.

Of the 325 observed animals, 322 were dogs and 3 were cats. The largest geographic share came from Uttar Badda with 219 observations, followed by Dokkhin Badda with 43, Moddho Badda with 39, and Jheelpar with 24.

Survey responses contained several free-text and inconsistent fields, particularly approximate weight and visible health issues. Weight entries included exact values, ranges, upper or lower bounds, textual descriptions, and values written using different formatting conventions. These entries were standardized where sufficient numerical information was available. Visible health descriptions were similarly normalized into broader analytical groups while preserving the original observations in the raw dataset.

Personally identifying information, including respondent contact details, addresses, and surveyor-related identifiers, was excluded from the cleaned analytical dataset. Image links were also not treated as clinical labels. Although photographs were collected for the surveyed animals, the present study does not use these images to train a computer-vision model or to make veterinary diagnoses.

The field survey is intentionally interpreted as a pilot case study rather than a representative survey of Bangladesh. Its purpose within the framework is to demonstrate how localized observations can complement broader Bangladesh-focused media intelligence. Consequently, percentages derived from the survey describe only the 325 observed animals and should not be interpreted as national prevalence estimates.

TABLE III
COMPOSITION OF THE BADDA PILOT FIELD SURVEY

Survey Attribute	Count
Total observed animals	325
Dogs	322
Cats	3
Uttar Badda	219
Dokkhin Badda	43
Moddho Badda	39
Jheelpar	24

G. Separation of the Two Evidence Streams

No row-level merge was performed between the news corpus and the field survey. Such a merge would be methodologically inappropriate because the two datasets describe different entities and were collected using different sampling processes. A news record represents a reported event at a particular time, whereas a field-survey row represents an animal observed within the Badda pilot area.

Instead, the two sources are integrated later at the analytical and decision-support level. News intelligence is used to examine reported incident patterns, while field intelligence is used to characterize localized welfare and intervention indicators. This separation is a central design decision of the proposed dual-source framework.

V. PROPOSED METHODOLOGY

The proposed framework was designed around two complementary sources of evidence: reported incidents from online news and direct observations from a localized field survey. Rather than forcing these heterogeneous sources into a single dataset, the system processes them through separate analytical branches and combines their outputs at the interpretation and decision-support stage.

The first branch, referred to as *News Intelligence*, performs collection, relevance screening, Bangladesh-scope refinement, incident-label construction, text representation, machine-learning classification, and SARI V2 assessment. The second branch, referred to as *Field Intelligence*, processes structured observations from the Badda pilot survey to examine health condition, vaccination, sterilization, age, behavior, and location-specific intervention indicators.

This architecture allows the broader Bangladesh-focused media evidence and the localized Badda observations to contribute different forms of information without implying that they were generated from the same sampling process.

A. Overall Dual-Source Architecture

Let the news collection be represented as

$$\mathcal{N} = \{n_1, n_2, \ldots, n_m\}, \tag{1}$$

where each n_i represents a retrieved news record containing available textual and metadata information. Similarly, the field-survey dataset is

$$\mathcal{F} = \{f_1, f_2, \ldots, f_k\}, \tag{2}$$

where each f_i represents one observed animal.

In the implemented study, the expanded news collection contained $m = 1094$ candidate records, while the Badda field dataset contained $k = 325$ animal observations.

The two branches can be represented conceptually as

$$I_N = g_N(\mathcal{N}), \tag{3}$$

and

$$I_F = g_F(\mathcal{F}), \tag{4}$$

where g_N denotes the news-intelligence pipeline and g_F denotes the field-analysis pipeline. Their analytical outputs, I_N and I_F, are interpreted jointly to form the overall welfare-intelligence output:

$$I_W = h(I_N, I_F). \tag{5}$$

Here, $h(\cdot)$ represents evidence-level integration rather than row-level data fusion. This distinction is necessary because a reported news event and an observed animal are not equivalent statistical units.

B. News Intelligence Pipeline

The news branch begins with the expanded candidate collection and applies a sequence of refinement operations. Duplicate removal is followed by content-based relevance filtering using the available title and RSS description. Geographic scope filtering then separates Bangladesh-primary records from foreign/background and out-of-scope material.

The sequence can be summarized as

$$1094 \rightarrow 690 \rightarrow 250 \rightarrow 226, \tag{6}$$

where 1,094 represents the expanded candidate collection, 690 the content-relevant corpus, 250 the Bangladesh-primary corpus, and 226 the final machine-learning corpus after incorporating assisted-review decisions.

The final text representation for each news record was constructed from the available textual content used by the processing pipeline. The resulting documents were converted into TF–IDF feature vectors before classification.

C. TF–IDF Text Representation

Raw text cannot be directly supplied to the selected conventional machine learning classifiers. The corpus was therefore transformed into a sparse numerical representation using Term Frequency–Inverse Document Frequency (TF–IDF).

For a term t in document d, term frequency may be expressed as

$$TF(t, d) = f_{t,d}, \tag{7}$$

where $f_{t,d}$ represents the occurrence of term t in document d. Inverse document frequency assigns greater importance to terms that occur in fewer documents:

$$IDF(t) = \log\left(\frac{N}{df(t)}\right), \tag{8}$$

where N is the number of documents and $df(t)$ is the number of documents containing term t. The corresponding TF–IDF weight is

$$TFIDF(t, d) = TF(t, d) \times IDF(t). \tag{9}$$

In practice, the implementation relies on the TF–IDF transformation provided by the machine-learning pipeline. The equations above describe the general weighting principle rather than claiming a separately implemented manual calculation.

TF–IDF was selected because the news corpus is relatively small and text representations are naturally high-dimensional and sparse. Under these conditions, linear models provide a practical baseline without requiring the substantially larger labeled datasets typically associated with data-intensive deep-learning approaches.

D. Incident Classification

Incident classification was formulated as an eight-class supervised text classification problem:

$$y \in \{A, C, D, H, K, O, P, R\}, \tag{10}$$

where the symbols represent Abuse, Conflict, Disease, Human-Animal Conflict, Killing, Other, Poisoning, and Rescue, respectively.

Three classifiers were evaluated under the same corpus and validation procedure: Linear Support Vector Machine, Logistic Regression, and Complement Naive Bayes.

1) Linear Support Vector Machine: A linear SVM seeks a separating hyperplane defined by

$$f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b, \tag{11}$$

where $\mathbf{x}$ is the TF–IDF feature vector, $\mathbf{w}$ is the learned weight vector, and b is the intercept. Linear SVMs are well suited to high-dimensional sparse text representations and have a long history of successful application to text categorization.

For the multiclass problem considered here, the implementation handles the extension beyond the binary formulation internally. The model is therefore used as an empirical text classifier rather than as a manually implemented optimization procedure.

2) Logistic Regression: Logistic Regression was included as a second linear classifier. For a binary case, the probability associated with a class can be expressed as

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + \exp[-(\mathbf{w}^T\mathbf{x} + b)]}. \tag{12}$$

The multiclass implementation extends this principle to the incident classification setting. Logistic Regression provides a

useful comparison with Linear SVM because both operate effectively on sparse text features but optimize different objectives.

3) Complement Naive Bayes: Complement Naive Bayes (CNB) was included because it was specifically developed to improve the behavior of multinomial Naive Bayes on imbalanced text-classification problems. This characteristic is relevant to the present dataset, where the largest class contains 90 records while the two rarest classes contain only three and two records.

CNB estimates class-discriminative information using statistics from the complement of each class. It therefore provides a probabilistic baseline with different assumptions from the two linear discriminative classifiers.

E. Cross-Validation Strategy

The final corpus contains substantial class imbalance. Most importantly, the Rescue category contains only two records. This imposed a practical constraint on the validation strategy because each validation split must preserve representation of the rare classes whenever stratification is used.

For this reason, model comparison was performed using stratified two-fold cross-validation. The choice of two folds was determined by the smallest class size rather than by an arbitrary preference for a low number of folds. A larger number of stratified folds would not be valid for a class represented by only two samples.

The same evaluation procedure was applied to all three candidate classifiers to maintain comparability.

F. Evaluation Metrics

Because the dataset is imbalanced, model selection was not based solely on accuracy. Precision, recall, and F1-score were considered alongside overall accuracy.

For a class of interest,

$$Precision = \frac{TP}{TP + FP}, \tag{13}$$

$$Recall = \frac{TP}{TP + FN}, \tag{14}$$

and

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall}. \tag{15}$$

Macro F1 assigns equal importance to each incident class:

$$F1_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} F1_c, \tag{16}$$

where $C = 8$ in the present study.

Weighted F1 instead weights each class by its support:

$$F1_{\text{weighted}} = \sum_{c=1}^{C} \frac{n_c}{N} F1_c. \tag{17}$$

Macro F1 was used as the principal model-selection criterion because a model that performs well only on the dominant classes would be undesirable for an incident-monitoring application. Weighted F1 and accuracy are reported as complementary measures.

G. Separation of Incident Classification and Risk Assessment

An important design decision in V2 was to avoid training a second supervised model to predict risk labels that were themselves deterministically generated from textual rules. Such an experiment could create circularity: a model might simply learn the same lexical indicators used to construct its target labels.

Instead, the supervised machine-learning component predicts *incident type*, while risk assessment is handled separately through SARI V2. Thus,

$$\text{Text} \rightarrow \text{Incident Classifier} \rightarrow \widehat{y}, \tag{18}$$

and independently,

$$\text{Text + Incident Context} \rightarrow \text{SARI V2}. \tag{19}$$

This separation improves interpretability and makes clear that classifier performance and risk scores answer different analytical questions.

H. SARI V2 Risk Assessment

The original SARI concept was revised in V2 to create an explicitly interpretable heuristic score ranging from 0 to 10. Five components are considered:

$$SARI_{V2} = A + H + S + P + C, \tag{20}$$

where

- A represents abuse or violence indicators, with a maximum contribution of 3.0;
- H represents human-conflict indicators, with a maximum contribution of 2.5;
- S represents sickness or rabies-related indicators, with a maximum contribution of 2.0;
- P represents poisoning indicators, with a maximum contribution of 1.5; and
- C represents contextual incident severity, with a maximum contribution of 1.0.

Therefore,

$$0 \leq SARI_{V2} \leq 10. \tag{21}$$

The contextual severity component assigns higher values to incident categories considered more immediately severe within the implemented heuristic. Poisoning and Killing receive a contextual contribution of 1.0, Abuse 0.9, Human-Animal Conflict 0.8, Disease 0.6, Conflict 0.4, Other 0.2, and Rescue 0.1.

The final score is converted into descriptive risk levels using

$$Risk = \begin{cases} Low, & SARI_{V2} < 3, \\ Medium, & 3 \le SARI_{V2} < 6, \\ High, & SARI_{V2} \ge 6. \end{cases} \tag{22}$$

These thresholds and component weights are heuristic design choices. They were not estimated from epidemiological outcomes or externally calibrated clinical data. SARI V2 should consequently be interpreted as an analytical prioritization index rather than as a probability of harm or a validated clinical risk score.

I. Field Intelligence Pipeline

The field-intelligence branch begins with the 325 animal observations from Badda. Raw survey responses were standardized before analysis. Categorical variables such as vaccination, sterilization, health condition, age group, and behavior were normalized, while inconsistent free-text health descriptions were grouped into broader analytical categories.

Approximate weight required additional preprocessing because surveyors used different formats, including exact values, numerical ranges, lower or upper bounds, and textual descriptions. When sufficient numerical information was available, the pipeline derived standardized minimum, maximum, or midpoint representations. Records without interpretable numerical weight information were not assigned fabricated values.

Three groups of field indicators were then examined:

1) **Animal characteristics**: animal type, age group, approximate weight, and behavior;
2) **Welfare indicators**: observed health condition and visible health issues; and
3) **Intervention indicators**: reported vaccination and sterilization status.

Location-specific summaries were generated for Uttar Badda, Dokkhin Badda, Moddho Badda, and Jheelpar. These summaries are descriptive of the observed sample only.

J. Nutritional Health-Concern Indicator

For summary analysis, an observed nutritional health-concern indicator was defined from the survey's health-condition field. Animals recorded as *Underweight / Malnourished* or *Starved* were grouped as showing a nutritional health concern.

If N_M is the number recorded as underweight or malnourished, N_S the number recorded as starved, and N_T the total number surveyed, the sample health-concern rate is

$$HCR = \frac{N_M + N_S}{N_T} \times 100. \tag{23}$$

In this study, $N_M = 65$, $N_S = 28$, and $N_T = 325$.

This indicator is based on field observation and should not be interpreted as a veterinary diagnosis or disease-prevalence estimate.

Algorithm 1 V2 Dual-Source Stray Animal Welfare Intelligence Framework

Require: Raw news records N, field survey records F
Ensure: Incident intelligence, SARI scores, field welfare indicators

Remove duplicate news records from N
Apply content-relevance filtering to N
Retain Bangladesh-primary news records
Generate candidate incident labels using domain rules
Apply assisted-review decisions to the reviewed sample
Construct the final ML corpus N^*
Convert news text in N^* to TF–IDF features
Evaluate Linear SVM, Logistic Regression, and Complement NB
Use stratified 2-fold cross-validation
Select the model with the highest Macro F1 score
Generate incident-category predictions
for all news records $n \in N^*$ **do**
    Compute separate SARI V2 components for n
    Calculate the SARI V2 score and risk level
end for
Clean and standardize field survey records F
Remove personally identifying information
Derive health, vaccination, sterilization, behavior, age, and location indicators
Calculate locality-level welfare summaries
Integrate news and field outputs at the evidence level
return incident intelligence, SARI V2 results, field intelligence, and decision-support indicators

K. Evidence-Level Integration

The final stage combines the interpretation of news and field outputs without merging their rows. News intelligence contributes information about reported incident categories, textual patterns, and heuristic SARI V2 severity. Field intelligence contributes information about the observed welfare and intervention conditions of animals in the pilot area.

The integrated framework therefore answers two related but distinct questions:

1) What types of stray-animal-related incidents are being reported in the Bangladesh-focused news corpus?
2) What welfare and intervention conditions are observable within the Badda pilot field sample?

Considering these questions together provides a broader basis for decision-support than either source can provide independently.

L. Framework Pseudocode

Algorithm 1 summarizes the complete V2 processing workflow. The two data sources are processed independently because a news record and an individual field observation represent different units of analysis. Their outputs are combined only at the evidence and decision-support level.

VI. Experimental Setup

A. Machine-Learning Corpus

Following relevance filtering, geographic scope refinement, and assisted review, the final machine-learning corpus contained 226 Bangladesh-primary news records. The classification task consisted of eight incident categories: Abuse, Conflict, Disease, Human-Animal Conflict, Killing, Other, Poisoning, and Rescue.

The corpus was strongly imbalanced. Other was the largest class with 90 records, whereas Poisoning and Rescue contained only three and two records, respectively. This imbalance influenced both the validation strategy and the choice of evaluation metrics.

B. Feature Representation and Candidate Models

News text was represented using TF–IDF features. Three conventional classifiers were evaluated:

- Linear Support Vector Machine (Linear SVM),
- Logistic Regression (LR), and
- Complement Naive Bayes (CNB).

These models were selected because they are computationally practical for a relatively small, high-dimensional, sparse text corpus. In particular, linear SVMs have been widely studied for text categorization [8], while Complement Naive Bayes was developed to address some of the weaknesses of standard Naive Bayes classifiers in imbalanced text-classification settings [9].

All models were evaluated using the same corpus and the same stratified two-fold cross-validation procedure. Two folds were used because the rarest class contained only two observations, making a larger number of stratified folds unsuitable.

C. Model Selection Criterion

Accuracy, Macro Precision, Macro Recall, Macro F1, and Weighted F1 were recorded for each candidate model. Macro F1 was used as the primary model-selection criterion because it gives equal importance to every class rather than allowing the dominant categories to determine the overall score.

This choice is particularly relevant in the present application. A classifier could achieve apparently strong overall accuracy by performing well on frequent classes such as Other while failing to recognize rare but potentially important categories such as Poisoning or Rescue. Therefore, accuracy and Weighted F1 were treated as complementary rather than sufficient measures of performance.

VII. Results and Analysis

A. Comparison of Incident Classification Models

Table IV presents the cross-validation results of the three candidate classifiers.

Linear SVM produced the highest Macro F1 score of 0.6068 and was therefore selected as the best-performing model. It also achieved the highest overall accuracy (0.7611), Macro Precision (0.7370), and Weighted F1 (0.7517). Logistic Regression produced a slightly higher Macro Recall of 0.5811, but its Macro F1 remained lower at 0.5944. Complement Naive Bayes achieved a Macro F1 of 0.5328.

The difference between Macro F1 and Weighted F1 is noteworthy. For Linear SVM, Weighted F1 reached 0.7517, while Macro F1 was 0.6068. This gap reflects the class imbalance in the corpus: performance on the more frequent categories contributes more strongly to the weighted metric, whereas Macro F1 exposes weaker performance on rare categories.

Figure 2 provides a visual comparison of the model results.

B. Class-Wise Performance of Linear SVM

A single aggregate score does not adequately describe the behavior of the selected model. Table V therefore reports class-level precision, recall, and F1-score for Linear SVM.

Disease achieved the highest F1-score (0.8684), followed by Abuse (0.8500) and Conflict (0.7869). The model also performed reasonably well on the dominant Other category, with an F1-score of 0.7644.

The results become less stable for classes with limited support. Human-Animal Conflict achieved a precision of 1.0000 but recall of only 0.2727. In other words, predictions assigned to this class were highly precise within the evaluated data, but many relevant records were not recovered. Poisoning shows a similar pattern, although its support of only three records makes the corresponding metrics particularly uncertain.

The Rescue category contained only two observations and obtained zero precision, recall, and F1-score. This result should not be interpreted as evidence that Rescue incidents are inherently impossible to classify. Rather, the available corpus does not contain enough examples for the model to learn a reliable decision boundary for this category.

Figure 3 shows the corresponding confusion matrix.

C. Interpretation of the Machine-Learning Results

The results suggest that conventional sparse-text models can recover useful structure from the refined news corpus, even with a relatively limited training set. At the same time, the experiment reveals an important data constraint: performance is closely related to the number and diversity of examples available for each incident category.

The selected model should therefore be viewed as a proof-of-concept incident classifier rather than a production-ready automated moderation or welfare assessment system. Increasing the quantity of carefully reviewed examples, particularly for Poisoning, Rescue, and Human-Animal Conflict, would be more valuable than interpreting small differences in model scores as evidence of complete model reliability.

There is an additional limitation in the evaluation procedure. Most labels in the corpus originated from rule-based weak supervision, followed by assisted review of a subset. The cross-validation results therefore measure how well the models reproduce patterns in this constructed labeling system. They do not constitute an independent expert-annotated benchmark. This distinction is retained throughout the interpretation of the results.

TABLE IV
CROSS-VALIDATION PERFORMANCE OF THE INCIDENT CLASSIFICATION MODELS

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Linear SVM	**0.7611**	**0.7370**	0.5722	**0.6068**	**0.7517**
Logistic Regression	0.7257	0.6792	**0.5811**	0.5944	0.7206
Complement Naive Bayes	0.7389	0.5949	0.5250	0.5328	0.7282

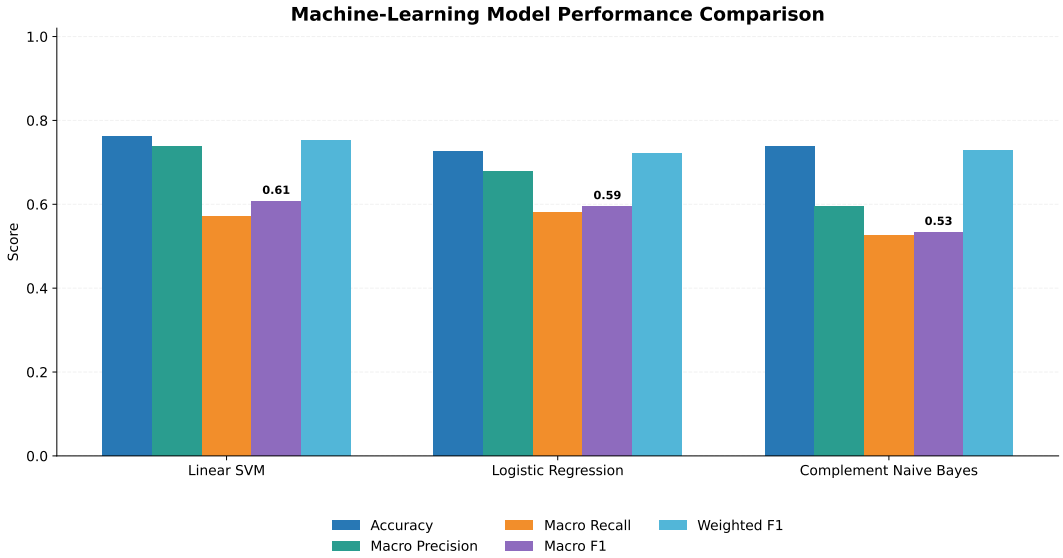


Fig. 2. Performance comparison of Linear SVM, Logistic Regression, and Complement Naive Bayes on the final incident-classification corpus.

TABLE V
CLASS-WISE PERFORMANCE OF THE SELECTED LINEAR SVM MODEL

Incident Category	Precision	Recall	F1-score	Support
Abuse	0.8947	0.8095	0.8500	21
Conflict	0.7500	0.8276	0.7869	29
Disease	0.8919	0.8462	0.8684	39
Human-Animal Conflict	1.0000	0.2727	0.4286	11
Killing	0.6364	0.6774	0.6562	31
Other	0.7228	0.8111	0.7644	90
Poisoning	1.0000	0.3333	0.5000	3
Rescue	0.0000	0.0000	0.0000	2

D. Incident Distribution in the Final Corpus

The final 226-record corpus remained dominated by the Other category, with 90 records. Disease accounted for 39 records, Killing for 31, Conflict for 29, and Abuse for 21. Human-Animal Conflict was represented by 11 records, while Poisoning and Rescue contained only three and two records, respectively.

Figure 4 illustrates this distribution.

This distribution provides useful context for interpreting the model performance. It also illustrates a practical challenge in media-derived datasets: some incident types are reported or captured far more frequently than others. The observed counts should therefore be interpreted as the composition of the collected news corpus, not as the true prevalence of stray-animal incidents across Bangladesh.

E. Temporal Distribution

The Bangladesh-primary news collection spans multiple years, with the final analytical corpus reflecting records available through the implemented collection process. Figure 5 presents the temporal distribution of the retained incidents.

The temporal pattern should be interpreted cautiously. The collection was constructed through retrospective news retrieval rather than a fixed, prospective surveillance system. Changes in news availability, search-engine indexing, source coverage, and reporting practices may therefore influence the number of records observed in different years.

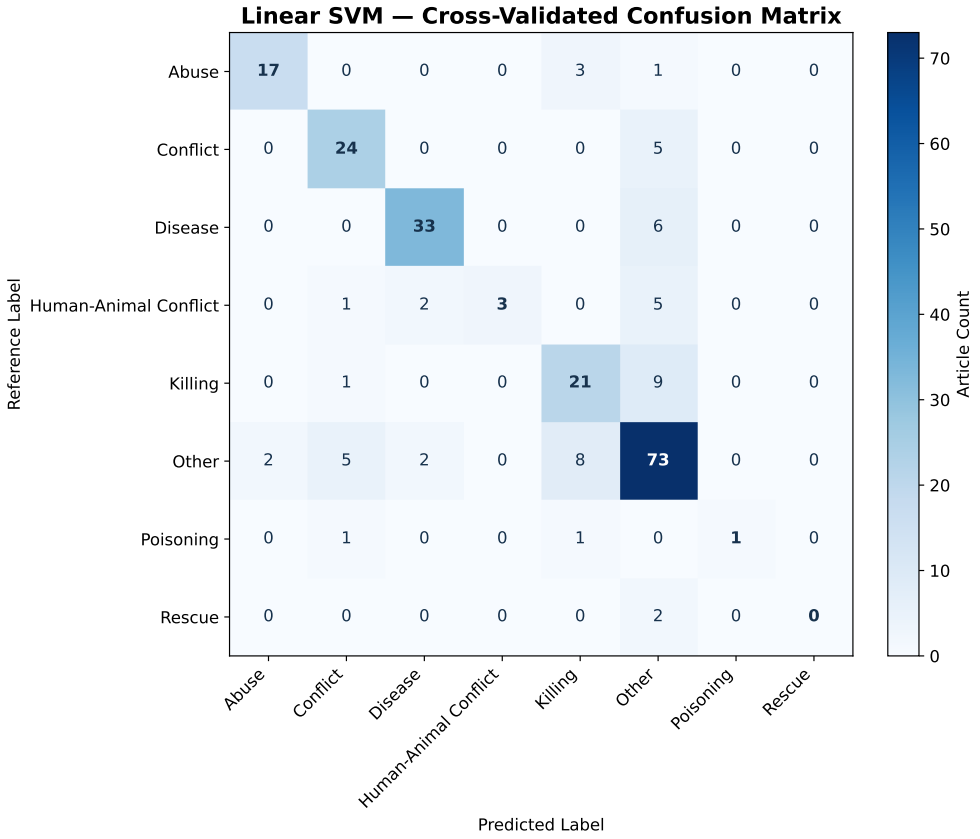


Fig. 3. Confusion matrix for the selected Linear SVM incident classifier.

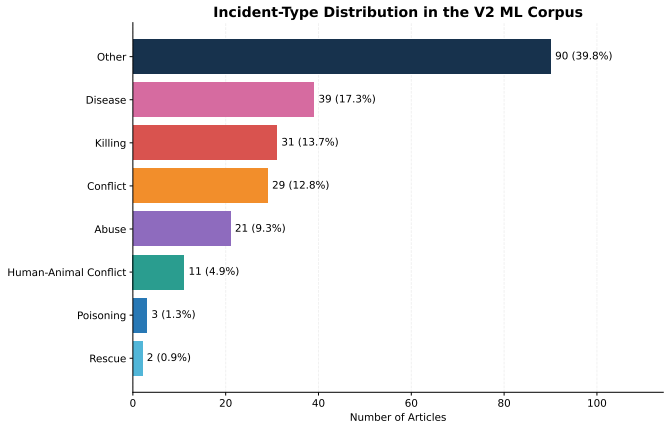


Fig. 4. Distribution of incident categories in the final 226-record machine-learning corpus.

TABLE VI
SUMMARY OF SARI V2 SCORES

Measure	Value
Number of records	226
Mean score	1.550
Standard deviation	1.344
Median	2.100
Minimum	0.100
Maximum	5.500
Low risk	191
Medium risk	35
High risk	0

In addition, records from 2026 represent only the period captured during data collection and should not be compared with completed calendar years as if they represented equivalent observation windows. For these reasons, the temporal visualization is used descriptively rather than as evidence of a causal increase or decrease in the underlying national incidence rate.

F. SARI V2 Risk Distribution

SARI V2 was calculated independently from the supervised model target. Across the 226 records, the mean score was 1.550 with a standard deviation of 1.344. The median was 2.100, and scores ranged from 0.100 to 5.500.

Using the predefined descriptive thresholds, 191 records were categorized as Low risk and 35 as Medium risk. No record reached the High-risk threshold of 6.0.

The absence of High-risk records is an actual outcome of the implemented scoring procedure and was retained without adjusting the thresholds to force a more visually balanced

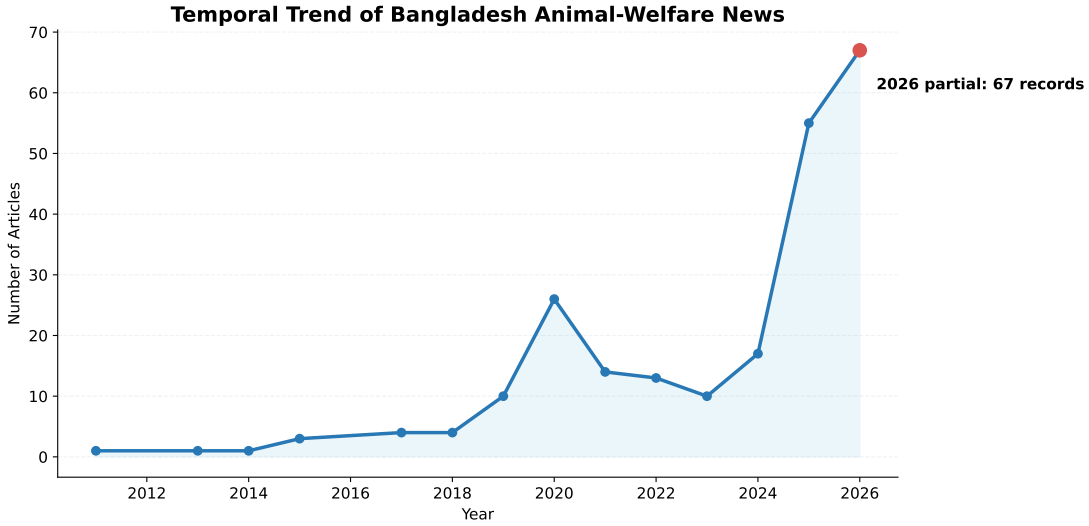


Fig. 5. Temporal distribution of incident records in the refined Bangladesh-focused news corpus.

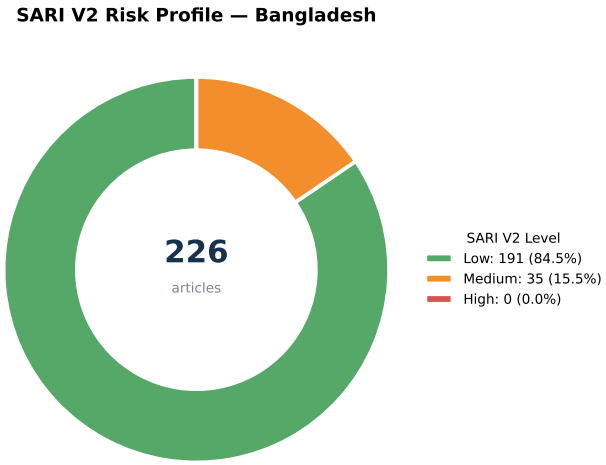


Fig. 6. Distribution of descriptive SARI V2 risk levels across the final news corpus.

distribution. The highest observed score was 5.500, which remained below the predefined High-risk boundary.

This finding also highlights the need to treat SARI V2 as an interpretable heuristic rather than a calibrated probability model. Its present value lies in ranking and summarizing textual risk indicators consistently; the weights and thresholds require external validation before the score could support high-stakes operational decisions.

G. SARI V2 by Incident Category

Average SARI V2 scores differed considerably across incident categories. Poisoning had the highest mean score at 4.833, followed by Human-Animal Conflict at 3.436. Abuse averaged 2.971, Killing 2.645, Disease 2.305, and Conflict 1.503. Other and Rescue produced substantially lower mean values of 0.222 and 0.100, respectively.

TABLE VII
MEAN SARI V2 SCORE BY INCIDENT CATEGORY

Incident Category	Mean SARI V2
Poisoning	4.833
Human-Animal Conflict	3.436
Abuse	2.971
Killing	2.645
Disease	2.305
Conflict	1.503
Other	0.222
Rescue	0.100

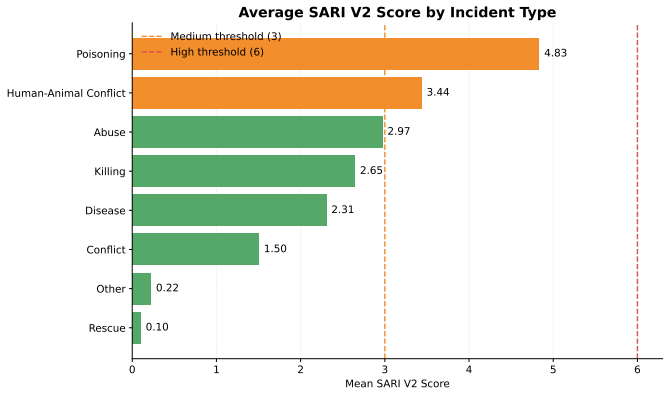


Fig. 7. Mean SARI V2 score across incident categories.

These differences are partly expected because incident context contributes directly to the SARI V2 heuristic. They should therefore not be interpreted as an independently discovered causal relationship between incident class and welfare risk. Instead, the comparison demonstrates how the implemented

TABLE VIII
SURVEY DISTRIBUTION BY LOCATION

Location	Observed Animals
Uttar Badda	219
Dokkhin Badda	43
Moddho Badda	39
Jheelpar	24
Total	**325**

risk logic prioritizes different combinations of textual and contextual signals.

VIII. FIELD SURVEY RESULTS

The field-survey component provides a localized view of stray-animal welfare conditions in Badda, Dhaka. A total of 325 animals were observed across Uttar Badda, Dokkhin Badda, Moddho Badda, and Jheelpar. Because the survey was not designed as a probability sample of Bangladesh, all percentages reported in this section refer only to the surveyed animals.

A. Animal Composition and Spatial Coverage

Dogs represented the overwhelming majority of observations, with 322 dogs and 3 cats. Uttar Badda accounted for 219 observations, followed by Dokkhin Badda with 43, Moddho Badda with 39, and Jheelpar with 24.

The strong concentration of observations in Uttar Badda indicates that the field sample is spatially uneven. This is important when interpreting location-specific rates, since local percentages can be influenced by the number of surveyed animals available in each area.

B. Observed Health Condition

The survey recorded three broad health-condition categories. Among the 325 animals, 232 were recorded as Healthy, 65 as Underweight / Malnourished, and 28 as Starved.

Combining the latter two categories gives 93 animals with an observed nutritional health concern. This corresponds to

$$\frac{65+28}{325} \times 100 = 28.62\%. \tag{24}$$

Therefore, approximately 28.62% of the surveyed animals showed an observed nutritional health concern according to the field-survey categories.

This measure should be interpreted cautiously. It reflects visual or field-level observations and is not equivalent to a veterinary diagnosis or an estimate of disease prevalence.

C. Vaccination Coverage

Only 32 of the 325 surveyed animals were recorded as vaccinated, while 293 were recorded as not vaccinated. The observed vaccination rate was therefore

$$Vaccination\ Rate = \frac{32}{325} \times 100 = 9.85\%. \tag{25}$$

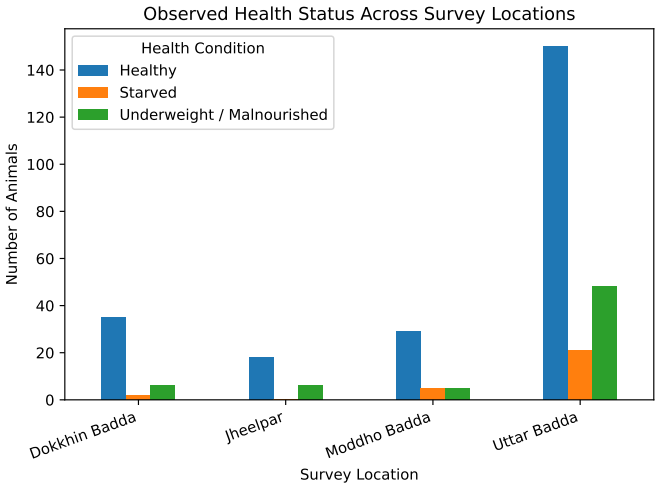


Fig. 8. Observed health-status distribution across the surveyed Badda locations.

The result indicates low vaccination coverage within the surveyed sample. Given the relevance of dog vaccination to rabies prevention, this represents a notable intervention gap within the pilot area. However, the result should not be generalized beyond the observed sample.

Location-wise analysis showed that no vaccinated animals were recorded among the surveyed animals in Dokkhin Badda. This statement refers specifically to the collected sample and does not imply that vaccination coverage is literally zero across the entire Dokkhin Badda area.

D. Sterilization Coverage

Sterilization coverage was similarly limited. Among the 325 animals, 34 were recorded as sterilized and 291 as not sterilized, corresponding to an observed sterilization rate of

$$Sterilization\ Rate = \frac{34}{325} \times 100 = 10.46\%. \tag{26}$$

Location-level analysis found no sterilized animals among the surveyed Jheelpar observations. As with vaccination, this finding is descriptive of the sampled animals and should not be interpreted as an area-wide prevalence estimate.

Figure 9 compares vaccination and sterilization coverage across the surveyed locations.

E. Location-Specific Welfare Concerns

The highest observed nutritional health-concern rate was found in Uttar Badda, where 31.51% of the surveyed animals were categorized as either Underweight / Malnourished or Starved.

Figure 10 illustrates the location-level distribution of this indicator.

The location-specific results should be interpreted jointly with the sample sizes. Uttar Badda has the largest number of observations and therefore dominates the overall survey

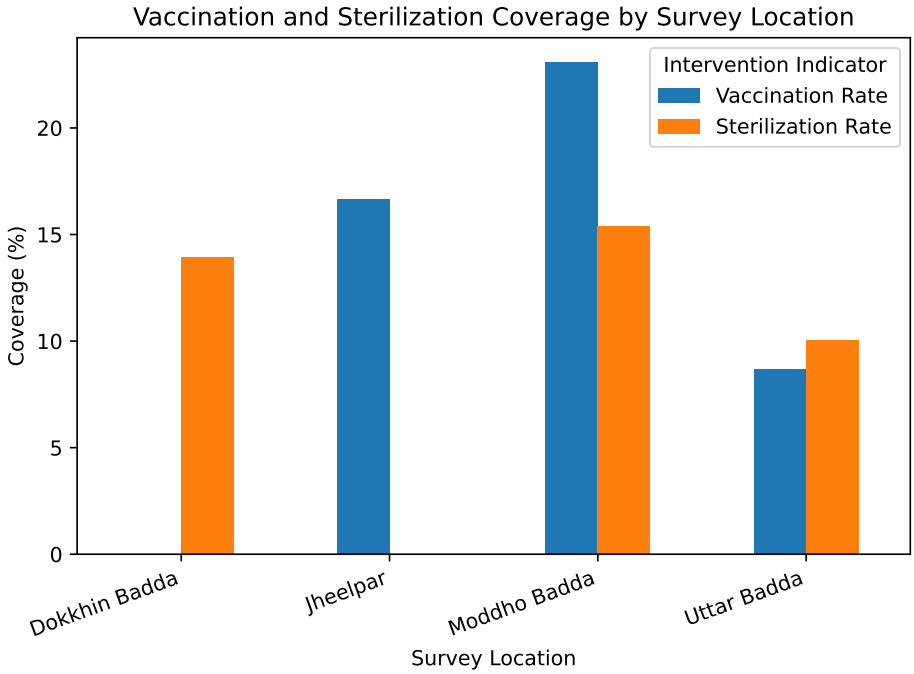


Fig. 9. Observed vaccination and sterilization coverage across the four Badda pilot locations.

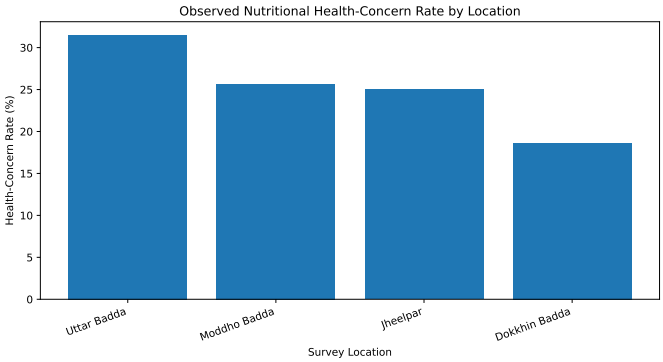


Fig. 10. Observed nutritional health-concern rate by survey location.

composition. Smaller areas such as Jheelpar have fewer observations, which makes their percentages more sensitive to small changes in the underlying counts.

F. Human-Directed Behavior

Most surveyed animals were recorded as Friendly toward humans. A total of 259 animals were categorized as Friendly, 50 as Scared, and 8 as Aggressive. A small number of records included combined behavioral descriptions.

This pattern suggests that overt aggression was relatively uncommon in the surveyed sample. Nevertheless, behavior recorded during a field observation can be influenced by the immediate environment, presence of people, food, injury, fear, or other situational factors. The survey therefore treats behavior as a descriptive field observation rather than a stable behavioral diagnosis.

G. Visible Health Issues

The raw survey included free-text descriptions of visible health problems. These descriptions varied substantially in wording and detail. During preprocessing, similar descriptions were grouped into broader categories such as skin-related issues, injury, mobility concerns, eye or ear problems, and other visible conditions.

The normalization process was used to support descriptive analysis while avoiding the assumption that surveyor observations constitute confirmed clinical diagnoses.

H. Field-Level Intervention Gap

The combination of low observed vaccination and sterilization coverage, together with the presence of animals showing nutritional health concerns, suggests that the pilot area contains identifiable intervention gaps.

Figure 11 summarizes these field-level indicators.

The purpose of this analysis is not to rank neighborhoods as "good" or "bad." Instead, the indicators illustrate how a local animal-welfare initiative could use structured field observations to prioritize follow-up surveying, vaccination campaigns, sterilization support, feeding assistance, or veterinary outreach.

IX. Integrated Welfare Intelligence

The central contribution of the proposed framework is the combination of two different but complementary forms of evidence. The news component captures reported incidents over a broader Bangladesh-focused information space, whereas the field component captures direct observations from a localized Badda pilot area.

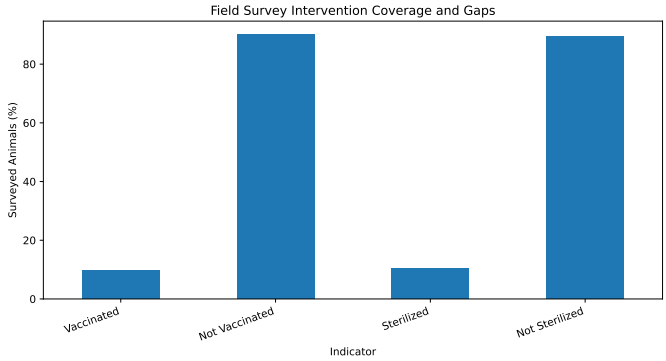


Fig. 11. Summary of observed intervention gaps in the Badda pilot field-survey sample.

These sources are not directly interchangeable. News reports are shaped by editorial selection, search visibility, public attention, and the severity or newsworthiness of events. Field surveys, in contrast, provide direct observations but are geographically limited and depend on the coverage of the survey team.

For this reason, the framework does not attempt to combine the two datasets into a single row-level model. Instead, their outputs are interpreted jointly to support welfare-oriented decision making.

A. News Intelligence Contribution

The news branch contributes three principal outputs:

- predicted incident categories from the selected Linear SVM model;
- descriptive incident and temporal patterns from the refined corpus;
- SARI V2 scores that provide an interpretable heuristic measure of textual and contextual severity.

Together, these outputs help organize a large set of unstructured news records into a more structured representation of reported animal-welfare events.

B. Field Intelligence Contribution

The field branch contributes direct observations related to animal condition and intervention coverage. In the Badda pilot sample, vaccination coverage was 9.85%, sterilization coverage was 10.46%, and 28.62% of animals were recorded as underweight/malnourished or starved.

These indicators address questions that the news corpus cannot answer reliably. News reports may describe severe or unusual events, but they do not provide systematic information on the routine vaccination, sterilization, or nutritional condition of free-roaming animals.

C. Complementary Interpretation

The two evidence streams therefore operate at different analytical levels. The news branch is event-oriented, while the field branch is condition-oriented.

This distinction can be summarized as

$$\textit{News Intelligence} \rightarrow \textit{Reported Events}, \tag{27}$$

$$\textit{Field Intelligence} \rightarrow \textit{Observed Conditions}. \tag{28}$$

The combined framework uses both perspectives to support a broader welfare assessment.

For example, a welfare organization may use incident intelligence to identify recurring categories of concern while using field observations to determine where vaccination, sterilization, nutritional support, or veterinary assessment may be most urgently needed.

D. Dual-Source Evidence Overview

Figure 12 presents a consolidated visual summary of the two evidence streams.

The integrated view should be interpreted as a decision-support framework, not as a national epidemiological surveillance system. The news corpus is not a census of all incidents in Bangladesh, and the field survey is not nationally representative. The framework instead demonstrates how two different forms of imperfect but useful evidence can be organized into a single analytical workflow.

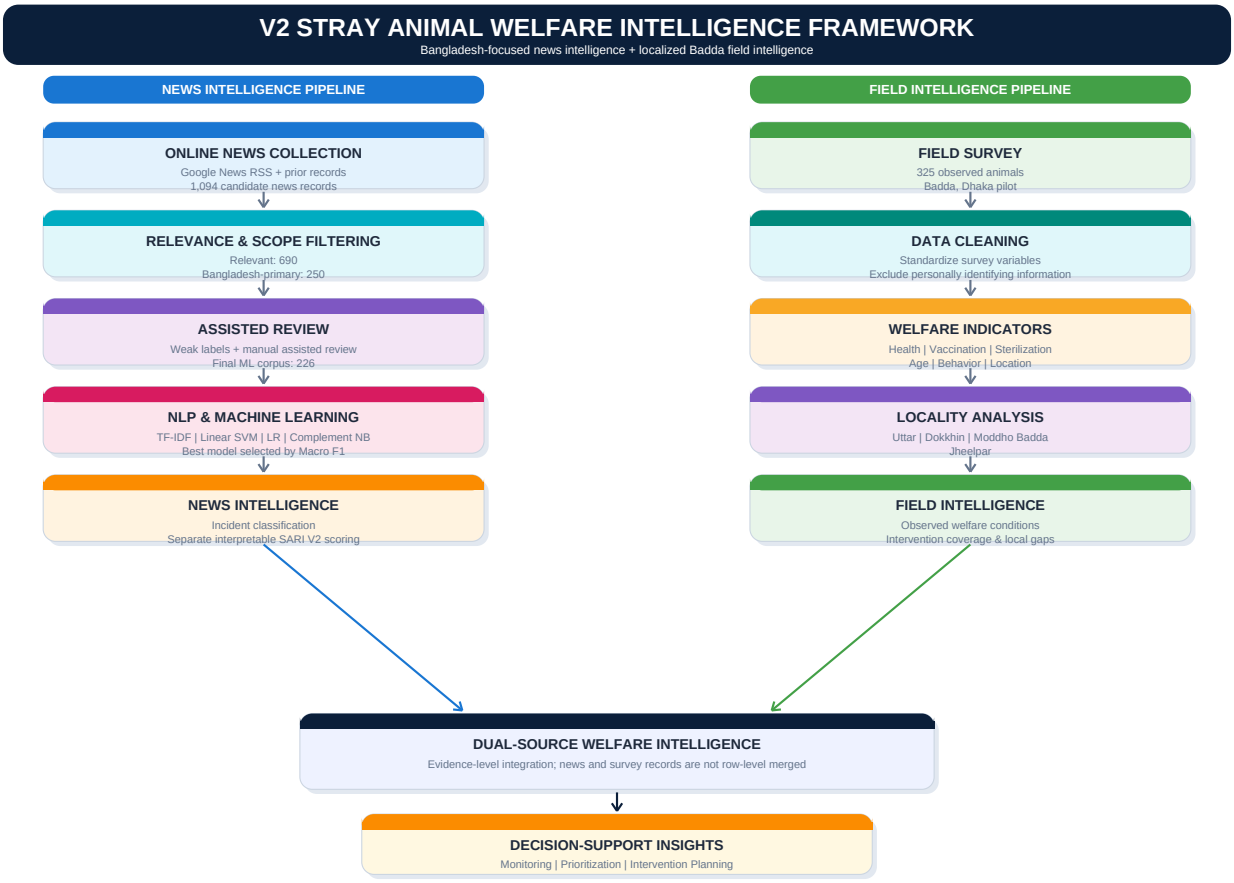


Fig. 12. Architecture of the proposed V2 stray animal welfare intelligence framework. The news branch performs relevance filtering, Bangladesh-specific scope refinement, assisted review, TF-IDF-based incident classification, and separate SARI V2 scoring. The field branch analyzes observed welfare indicators from the Badda pilot survey. Both branches are integrated at the evidence level to support monitoring, prioritization, and intervention planning.

X. Model Interpretability

Predictive performance alone provides limited value in a welfare-intelligence system if the basis of a model's decisions cannot be inspected. For this reason, the selected Linear SVM was examined through its learned feature weights. Since the classifier operates on TF–IDF representations, terms with larger positive coefficients for a particular class provide an indication of which textual features most strongly influence that class within the fitted model.

The analysis was performed on the trained Linear SVM stored after model selection. For each incident category, the highest-weighted TF–IDF features were extracted and ranked. This provides a lightweight and model-specific form of interpretability without introducing a separate explanation model.

Figure 13 presents the most influential textual features identified for the incident classes.

The feature analysis is useful for two reasons. First, it allows inspection of whether the classifier is learning terms that are reasonably connected with the intended incident categories. Second, it can reveal undesirable shortcuts, such as excessive dependence on a source name, location, or a small number of highly specific keywords.

However, the feature weights should not be interpreted as causal explanations. A high coefficient indicates that a feature contributes strongly to a decision boundary learned from this corpus; it does not imply that the term causes the underlying welfare event.

A further concern arises from the labeling procedure. Most training labels were initially generated using domain-specific textual rules. Some of the same lexical signals may therefore appear both in the weak-label construction process and in the TF–IDF representation used by the classifier. This can produce a degree of label-feature dependence and may make classification performance appear stronger than it would be against a fully independent expert-annotated dataset. The feature analysis consequently serves not only as an explanation mechanism but also as a diagnostic tool for identifying this potential source of leakage.

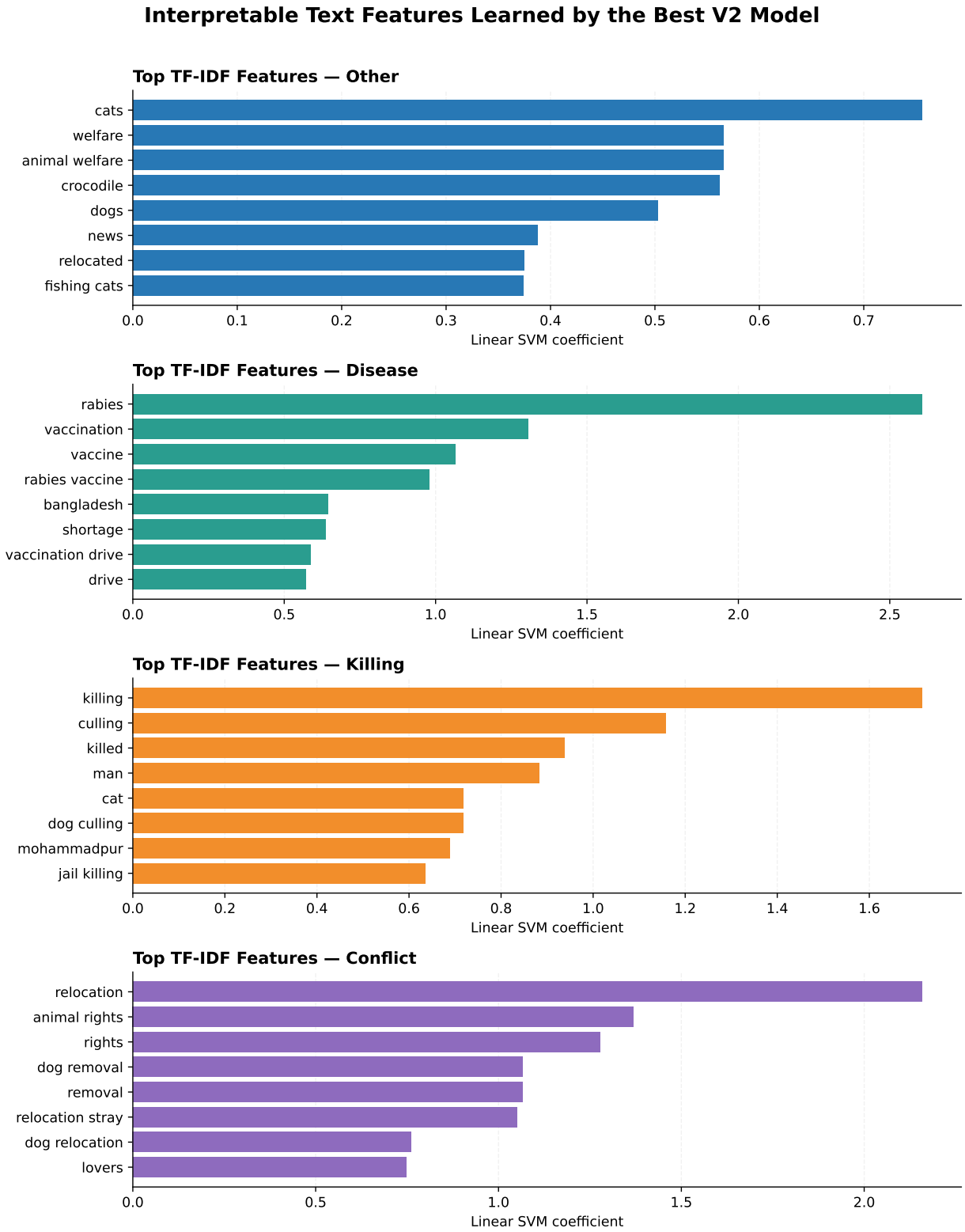


Fig. 13. Representative highest-weighted TF–IDF features for four incident categories identified by the selected Linear SVM model. Positive feature weights indicate terms that contributed more strongly to the corresponding class decision.

XI. Evolution from the Earlier Framework

The present work is an extension of an earlier news-based stray-animal risk assessment pipeline. The earlier version established the feasibility of collecting online news records, assigning rule-based incident categories, calculating a simple

severity score, and visualizing the resulting patterns. That version provided a useful proof of concept, but several limitations became apparent during its implementation.

In particular, incident assignment relied primarily on predefined lexical rules, textual context was limited, geographic information was inconsistent, and the risk calculation was simplified in implementation. The earlier framework also lacked a direct field-observation component.

V2 was therefore designed as a methodological extension rather than merely an increase in dataset size.

The comparison demonstrates that the main advancement in V2 is not a single classifier or visualization. The larger change is the restructuring of the project into a more defensible analytical workflow in which data scope, label quality, model evaluation, risk assessment, and field evidence are treated as separate methodological problems.

XII. Discussion and Practical Implications

The findings illustrate both the potential and the limitations of using heterogeneous data for stray-animal welfare analysis in Bangladesh. The news pipeline shows that publicly available reports can be transformed into a structured incident corpus, while the field survey demonstrates that important welfare indicators are often visible only through direct local observation.

A. Lessons from the News Intelligence Component

One of the clearest findings from the news pipeline is the importance of data refinement. Starting from 1,094 candidate records, only 690 remained after content relevance filtering, and only 250 were classified as Bangladesh-primary. Following assisted-review decisions, the final machine-learning corpus contained 226 records.

This reduction is not simply data loss. It reflects an attempt to reduce noise introduced by search retrieval, foreign reports, wildlife-related content, and ambiguous uses of animal-related terminology. A larger but poorly scoped dataset would have produced more observations while weakening the connection between the analysis and the intended Bangladesh context.

The Linear SVM achieved the strongest overall balance among the evaluated classifiers, with a Macro F1 of 0.6068 and Weighted F1 of 0.7517. These results are encouraging for a proof-of-concept system, particularly given the small corpus. At the same time, the failure to recover the Rescue class and the weak recall for several rare categories demonstrate that the current dataset is not sufficient for reliable automated classification across every incident type.

B. Lessons from SARI V2

SARI V2 adds a different analytical perspective. Rather than attempting to predict a weakly constructed risk label through another supervised model, it uses transparent components to summarize the textual and contextual signals associated with an incident.

The resulting scores were concentrated in the Low and Medium ranges, with no record reaching the predefined High-risk threshold. This result is useful precisely because the scoring thresholds were not changed after observing the distribution. Adjusting the threshold merely to generate High-risk cases would make the resulting visualization more dramatic but would weaken the methodological consistency of the framework.

The current SARI V2 should therefore be viewed as a prototype prioritization mechanism. Future validation against independently assessed welfare outcomes would be necessary before assigning operational meaning to particular score thresholds.

C. Lessons from the Badda Pilot Survey

The field survey reveals a different set of concerns. Among 325 observed animals, only 9.85% were recorded as vaccinated and 10.46% as sterilized. In addition, 28.62% were recorded as either underweight/malnourished or starved.

These results do not establish prevalence for Badda as a whole, and they certainly do not represent Bangladesh nationally. Nevertheless, they demonstrate the type of local information that structured welfare surveys can produce. Such information can support practical planning for vaccination, sterilization, feeding assistance, veterinary assessment, and repeat surveying.

The location-specific results can also be used as prompts for further investigation. For example, no vaccinated animals were observed within the surveyed Dokkhin Badda sample, while no sterilized animals were recorded within the surveyed Jheelpar sample. These findings should trigger additional data collection rather than immediate area-wide conclusions.

D. Potential Use in Welfare Operations

A future operational version of the framework could support local animal welfare groups, researchers, and relevant authorities through a shared decision-support interface. Incoming news could be organized into incident categories, while field teams could update structured animal observations from selected localities.

The resulting system could help answer practical questions such as which incident categories are appearing in the monitored media corpus, which locations have low observed intervention coverage, where repeat surveys may be useful, and which records deserve manual review.

Human review would remain important. The framework is intended to organize evidence and support prioritization, not to automate veterinary diagnosis, legal judgment, or intervention decisions.

XIII. Limitations

Several limitations should be considered when interpreting the results.

First, the news dataset is based on Google News RSS retrieval and therefore does not represent a complete census of stray-animal incidents in Bangladesh. Search ranking, indexing, source availability, query design, and media attention all influence which events appear in the collected corpus.

Second, full article text was not consistently available. The analysis relies primarily on headlines and available RSS

TABLE IX
METHODOLOGICAL EVOLUTION FROM V1 TO V2

Component	Earlier Framework (V1)	Current Framework (V2)
News processing	News scraping followed by basic cleaning and rule-based categorization.	Expanded retrieval followed by content relevance filtering, Bangladesh-scope refinement, and assisted review.
Incident analysis	Incident categories assigned using deterministic textual rules.	Eight-class supervised text classification using TF–IDF with Linear SVM, Logistic Regression, and Complement Naive Bayes.
Model evaluation	No comparative supervised incident-classification experiment.	Stratified cross-validation with Accuracy, Macro Precision, Macro Recall, Macro F1, and Weighted F1.
Risk assessment	Simple rule-based severity assignment associated closely with incident type.	Separate five-component SARI V2 heuristic with explicit 0–10 scoring and documented thresholds.
Interpretability	Primarily rule-level transparency.	Inspection of learned Linear SVM TF–IDF feature weights in addition to transparent SARI components.
Geographic handling	Location information was frequently unavailable or inconsistent.	Explicit Bangladesh-primary scope refinement; unsupported geographic hotspot claims are avoided.
Field evidence	Not incorporated.	Structured pilot observations of 325 free-roaming animals across four Badda localities.
Framework design	Primarily news-oriented analytical pipeline.	Dual-source architecture combining event-oriented news intelligence with condition-oriented field intelligence.

descriptions. This restricts the amount of contextual information available for relevance filtering, incident classification, geographic interpretation, and SARI V2 scoring.

Third, the incident labels are not a fully independent expert-annotated gold standard. They were generated through weak supervision and partially checked through assisted review. Although the reviewed retained subset showed 90.0% agreement with the initial suggested labels, this value is not an estimate of model accuracy or expert annotation quality.

Fourth, lexical rules used during weak-label construction can overlap with features later learned by the TF–IDF classifier. This creates a potential source of label-feature dependence. Consequently, the reported cross-validation scores may overestimate performance relative to evaluation against independently annotated unseen data.

Fifth, the final machine-learning corpus is small and strongly imbalanced. Poisoning contains only three records and Rescue only two. Metrics for these classes are therefore unstable, and the selected classifier should not be considered production-ready.

Sixth, SARI V2 is a heuristic index. Its component weights and Low, Medium, and High thresholds were designed for interpretability but have not been externally calibrated against clinical, epidemiological, or independently assessed welfare outcomes.

Seventh, geographic analysis of the news corpus remains limited. Reliable coordinates were not available for the majority of records, so techniques such as DBSCAN or HDBSCAN hotspot detection were not applied. Introducing such methods without reliable location data would create artificial precision.

Eighth, the Badda field survey is a localized pilot sample. Survey coverage is uneven across the four localities, with substantially more observations from Uttar Badda. The sample was not designed to estimate population-level prevalence, and the observed health categories were not confirmed through veterinary examinations.

Finally, although photographs were collected during the field survey, they were not used as verified labels for computer-vision training or clinical assessment. The current study therefore makes no image-based diagnostic claims.

XIV. FUTURE WORK

The next stage of the project should focus primarily on improving evidence quality rather than simply increasing model complexity.

A larger independently annotated news corpus would allow more reliable model evaluation and provide sufficient examples for rare categories such as Poisoning, Rescue, and Human-Animal Conflict. Annotation by multiple human reviewers would also make it possible to measure inter-annotator agreement and construct a stronger evaluation benchmark.

The news pipeline can be extended from headlines and RSS descriptions to full-text article processing where legally and technically available. Richer text would support more accurate event extraction, location identification, entity recognition, and contextual classification.

Geographic analysis is another important direction. If reliable coordinates or consistently normalized location fields become available, the framework could support spatial clustering and hotspot analysis. Methods such as DBSCAN or HDBSCAN should be introduced only after the geographic evidence is sufficiently complete and accurate.

SARI V2 also requires further validation. Future work could involve animal welfare professionals or veterinarians in reviewing representative incidents and assessing whether the component weights and thresholds correspond to meaningful intervention priorities.

The field component should be expanded beyond Badda using a more systematic sampling strategy. Repeated surveys across additional neighborhoods and districts would allow the framework to distinguish persistent conditions from short-term

observations and would improve the geographic relevance of the field intelligence.

Survey photographs could eventually support computer-vision research, but this would require explicit image-label definitions, quality control, privacy safeguards, and substantially more verified examples. Image-based health assessment should not be attempted solely from the existing unvalidated photographs.

Finally, the two evidence streams could be exposed through an interactive dashboard with appropriate uncertainty indicators. Such a platform could support incident review, survey monitoring, intervention tracking, and periodic updates while retaining human oversight.

XV. Conclusion

This study presents a machine learning-driven stray-animal welfare intelligence framework designed for the Bangladesh context. The work extends an earlier news-based risk-assessment pipeline by introducing stricter data refinement, supervised incident classification, an interpretable SARI V2 risk index, and a localized field-intelligence component.

The news pipeline expanded to 1,094 candidate records, of which 690 remained after relevance filtering, 250 were identified as Bangladesh-primary, and 226 formed the final machine-learning corpus after assisted-review decisions. Among three evaluated classifiers, Linear SVM produced the strongest Macro F1 score of 0.6068 and a Weighted F1 of 0.7517.

SARI V2 provided a separate heuristic assessment of textual and contextual severity. The mean score was 1.550, with 191 records categorized as Low and 35 as Medium; no record crossed the predefined High-risk threshold.

The field component added direct observations of 325 free-roaming animals from four localities in Badda, Dhaka. Within this pilot sample, 9.85% were recorded as vaccinated, 10.46% as sterilized, and 28.62% showed an observed nutritional health concern.

The main contribution of the study lies in connecting these two perspectives without treating them as equivalent datasets. News intelligence provides a view of reported events, while field intelligence provides a view of observed local conditions. Their evidence-level integration creates a more complete foundation for welfare-oriented decision support than either source could provide independently.

The framework remains a proof of concept rather than a national surveillance or clinical decision system. Nevertheless, it establishes a reproducible foundation that can be strengthened through expert annotation, broader field coverage, improved geographic information, validated risk scoring, and continued data collection.

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